Write Up

How Well Does Steering Demonstrate a Model's True Diversity?

Matthew Khoriaty, David Williams-King, Shi Feng

Abstract

Activation steering approaches have become an important part of an AI safety researcher's toolkit due to its flexibility in targeting properties of interest. However, there has not been a study into what this does to the model's diversity of output. This research investigates the effect that steering methods have on a model's output diversity. The overarching goal is to check the range of outputs the steering method is capable of producing. In particular, this research will focus on eval nonawareness steering. If the unaware model has limited output diversity, then that indicates that steering models to be unaware is not a robust way to test models for concerning output in deployment settings.

Prior Work

- Liars' Bench: Evaluating Lie Detectors for Language Models
- Why Steering Works: Toward a Unified View of Language Model Parameter Dynamics
- https://forum.effectivealtruism.org/posts/rAL3YAYYr6gGcyzq5/thinking-about-propensity-evaluations
- Analyzing the Generalization and Reliability of Steering Vectors
- Claude on Clustering Approaches
- Claude on this doc 2026-02-18
- Understanding the effects of RLHF on LLM generalisation and diversity
    - They got their evals from Evaluating the Evaluation of Diversity in Natural Language Generation
- Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs
- Steering Evaluation-Aware Models to Act Like They Are Deployed
- The Hawthorne Effect in Reasoning Models: Evaluating and Steering Test Awareness
- The Truthfulness Spectrum Hypothesis
    - Domain-specific steering works better than general steering. The paper finds a good general steering vector by aggregation of specific ones.
    - Matthew is unimpressed. It doesn't seem relevant, and there probably isn't transfer because sentence embedding models are already optimized to make cosine similarity work.
- https://openreview.net/forum?id=2U1KIfmaU9

    - Claude analysis of this plus my work https://claude.ai/share/c1a2232b-c166-4f7b-bcab-b1c8c06495dc

Methods

I identified eval-aware open-source models using the evaluation from https://github.com/jjpn97/eval_awareness.

Run sentence embedding for diversity.
Run n-gram experiments for diversity.
Develop mutual-information-based diversity metric.
Pass@k for MTBench. Pass@k requires multiple questions with answers that can be evaluated, so it measures something different from "take many responses from a single prompt and find the diversity". Can test the steering differences for pass@k on dangerous or dangerous adjacent dataset.

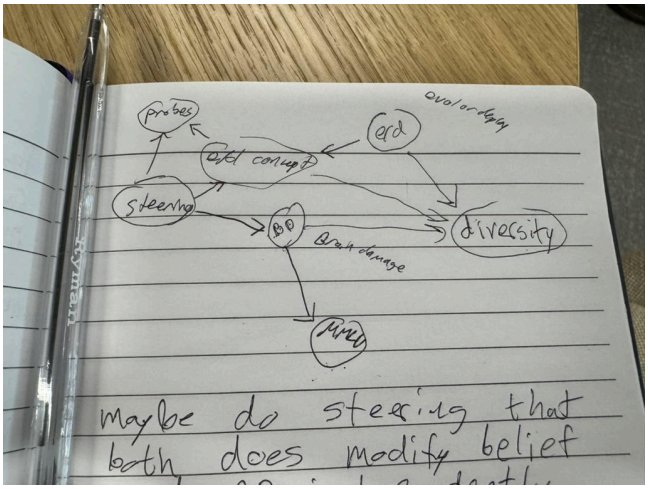
Inject within-prompt diversity by making subtle changes/paraphrases?
Test the eval non aware models for if they can identify they are being steered and to what concept. I predict they don't.

Experiment 1: Simple Diversity Experiment

- We want to evaluate the output diversity of steered models.
- Steering vectors and infrastructure for training them can be found on EasySteer Github.
    - Using pre-existing vectors reduces chance for p-hacking
- euclaise/writingprompts is a fun dataset of prompts we can have models respond to.
- We need a clustering method and diversity metric
    - Sentence-BERT embedding cosine similarity
    - HDBSCAN for clusters

Future:

- Run LM as clusterer for diversity. Also ask for an "intelligbility score" from 0-5. Few shot prompting tends to work well,.Can use a real example of intelligibility drop as an intelligibility 1. Can use "..." to save tokens.
- Make plot of every pairwise cosine distance
- More values for the diversity graphs
    - More prompts, more samples per prompt, more subdivisions, especially in the 0-4 range.
- Hunt bugs in Figure 1.
- Unfortunately, the most eval-aware models are huge. I'll use Qwen3-30B-A3B with its measly .65 AUC because it has the best size/eval-awareness tradeoff.

- Test Neuronpedia's steering vector on the diversity benchmark
    - Neuronpedia's steering vector
        - "The most popular"
        - Neuronpedia only has SAEs
- Measure 'coherence'. Can use prompted LM judge or KL divergence
- Cook's distance on https://github.com/AMindToThink/icl-diversity/blob/main/investigations/cross_mode_surprise_drop/figures/pairwise_row_vs_col.png
- Steering methods to try:
    - Try whatever steering method Anthropic uses
    - Contrastive Activation Addition
    - SAEs
    - LoReft
    - LoRRA (technique by Centre for AI Safety)
- Causal considerations?
    - What if the "deployment" concept triggers diversity because "evals" normally have "one correct answer"?
        - Prompted baseline? Deployment data + (deploy or eval or nothing) prompt, measure diversity to get the effect of the "concept" of eval/deployment on diversity
    - 

    - Compare against random vectors
        - Would this be informative? I would expect semantically meaningful model-extracted vectors to behave in a qualitatively different way than random ones.
    - Could we do steering that both modifies belief in eval/deployment and model degradation independently?

Results

Diversity

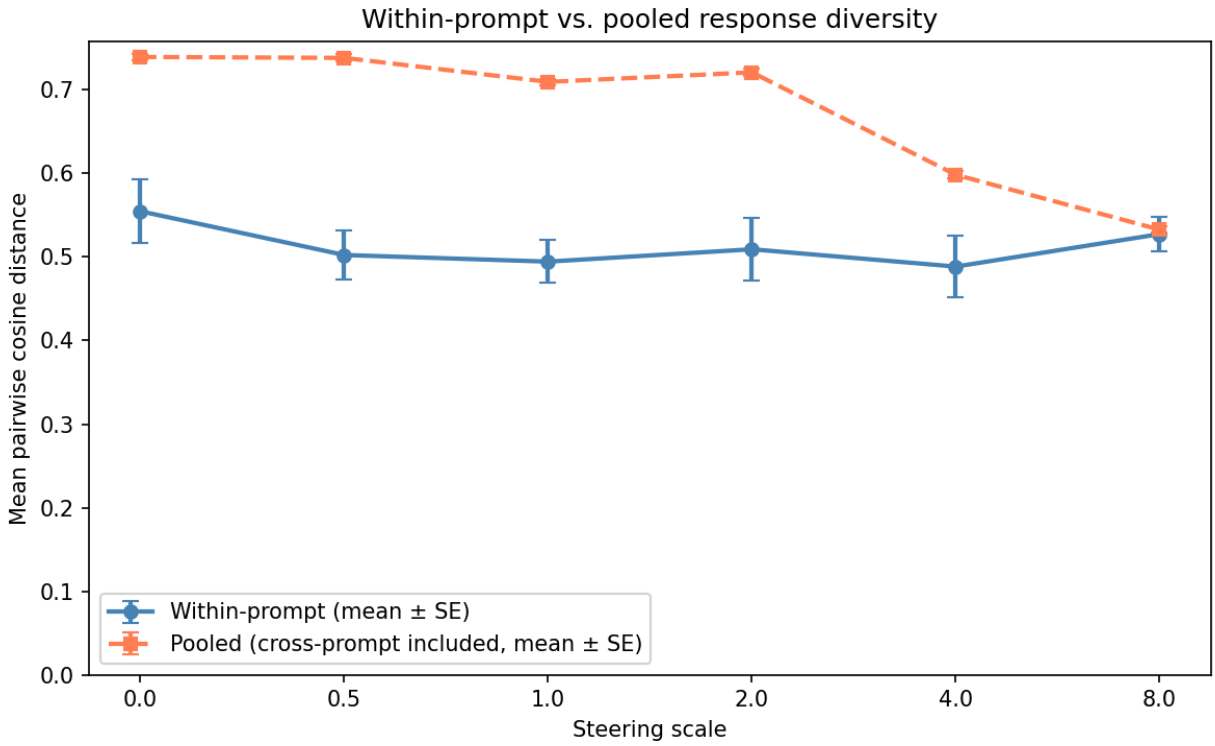
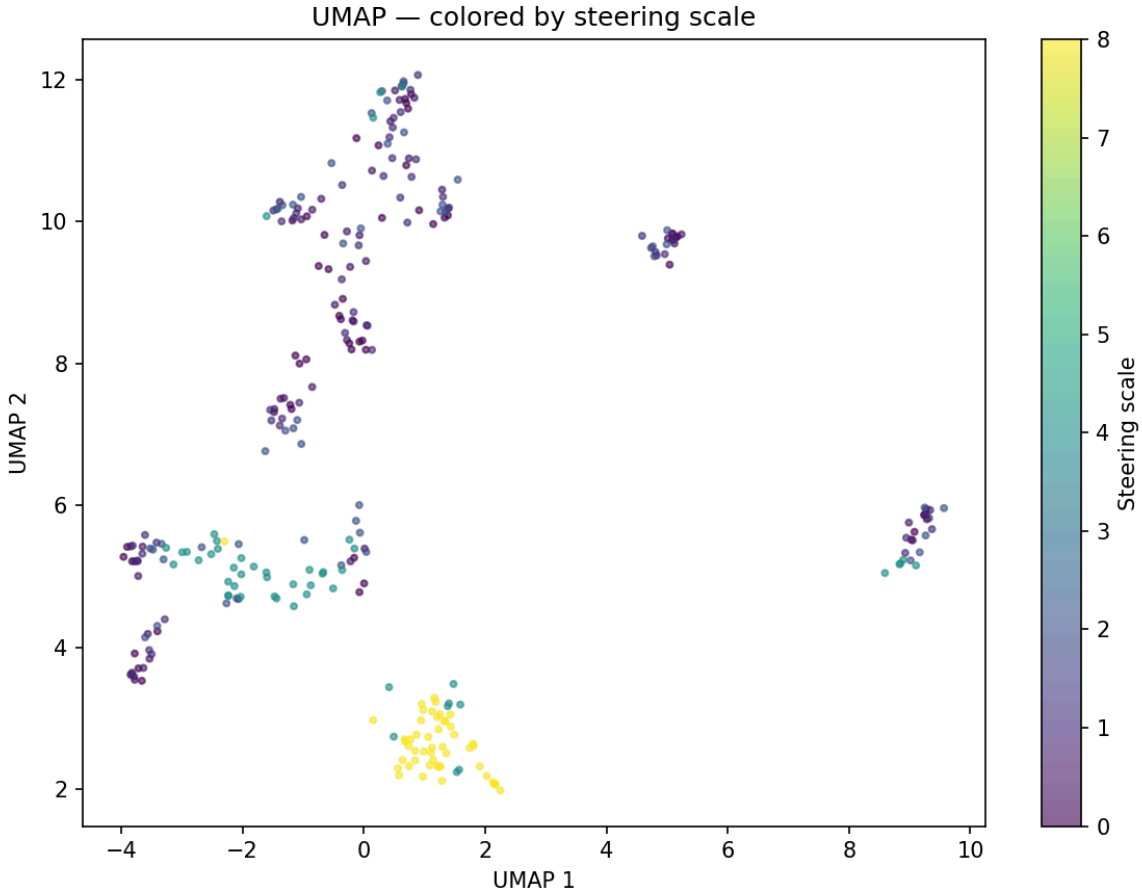


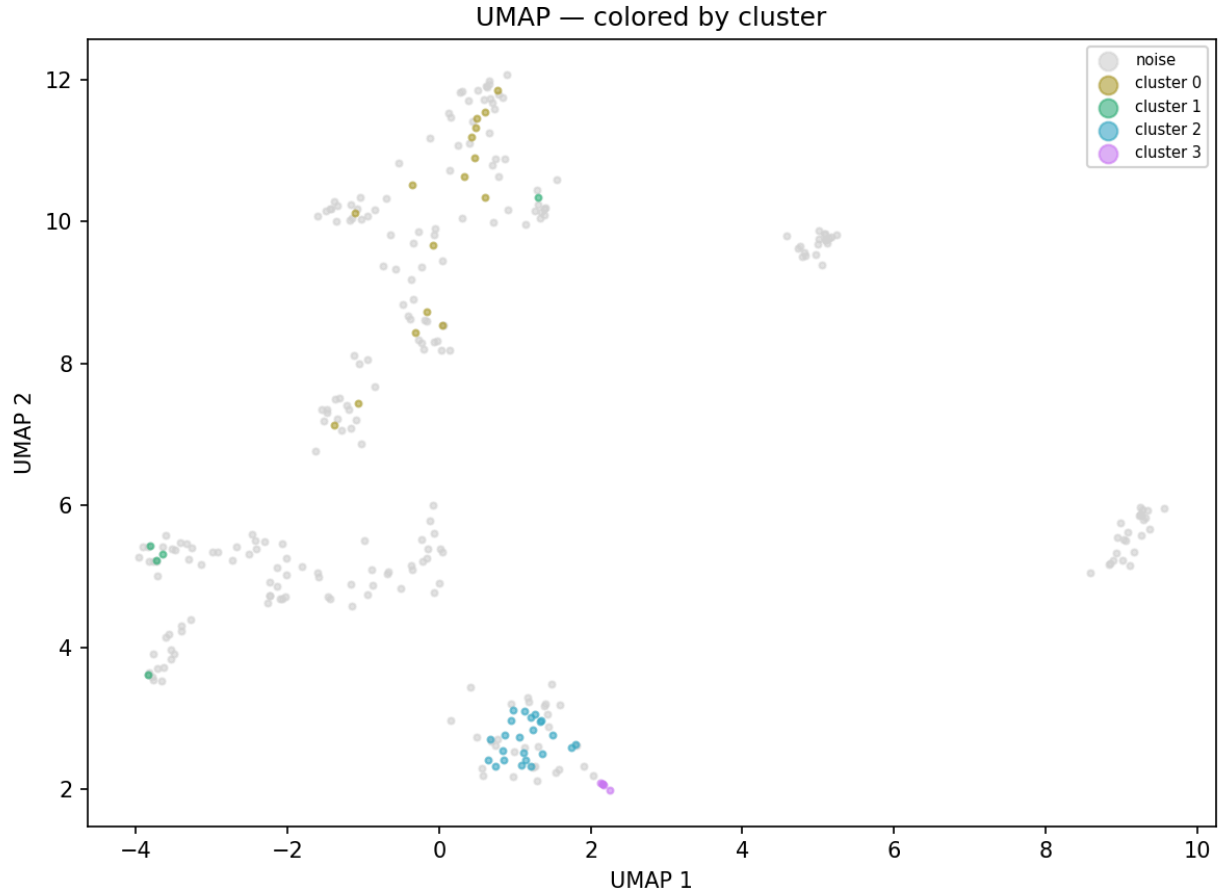
Figure 1: This is odd: within prompt diversity did not change with steering scale according to mean pairwise cosine distance.

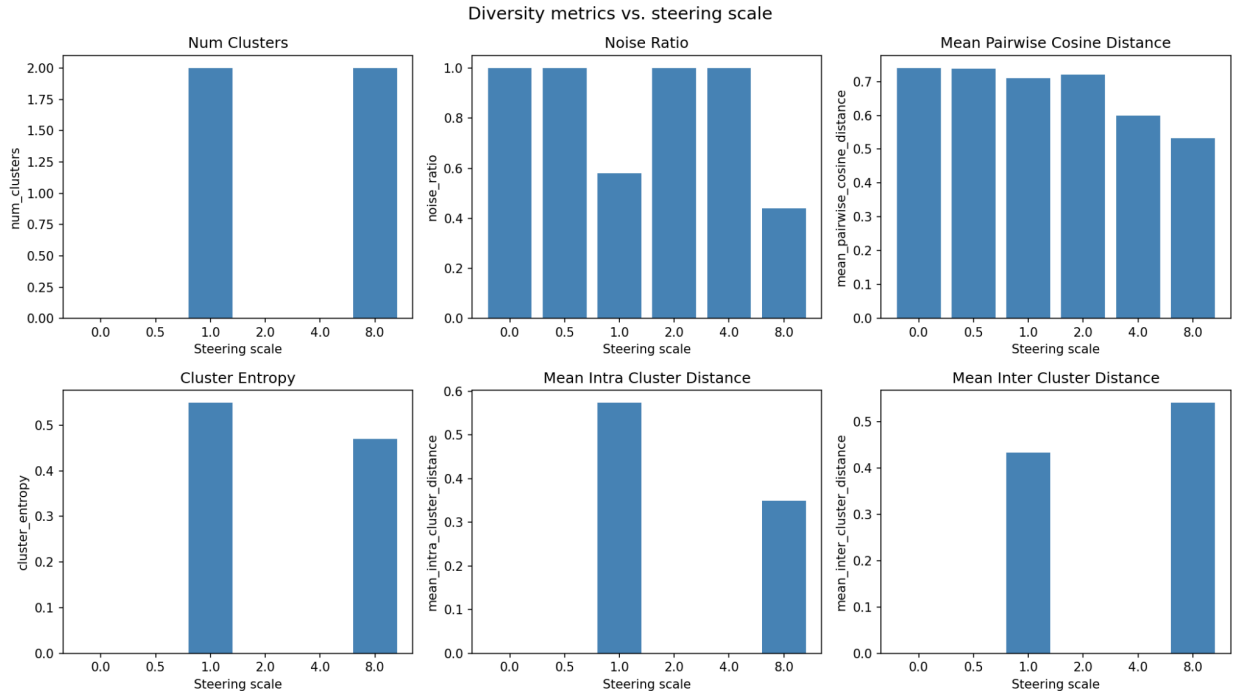


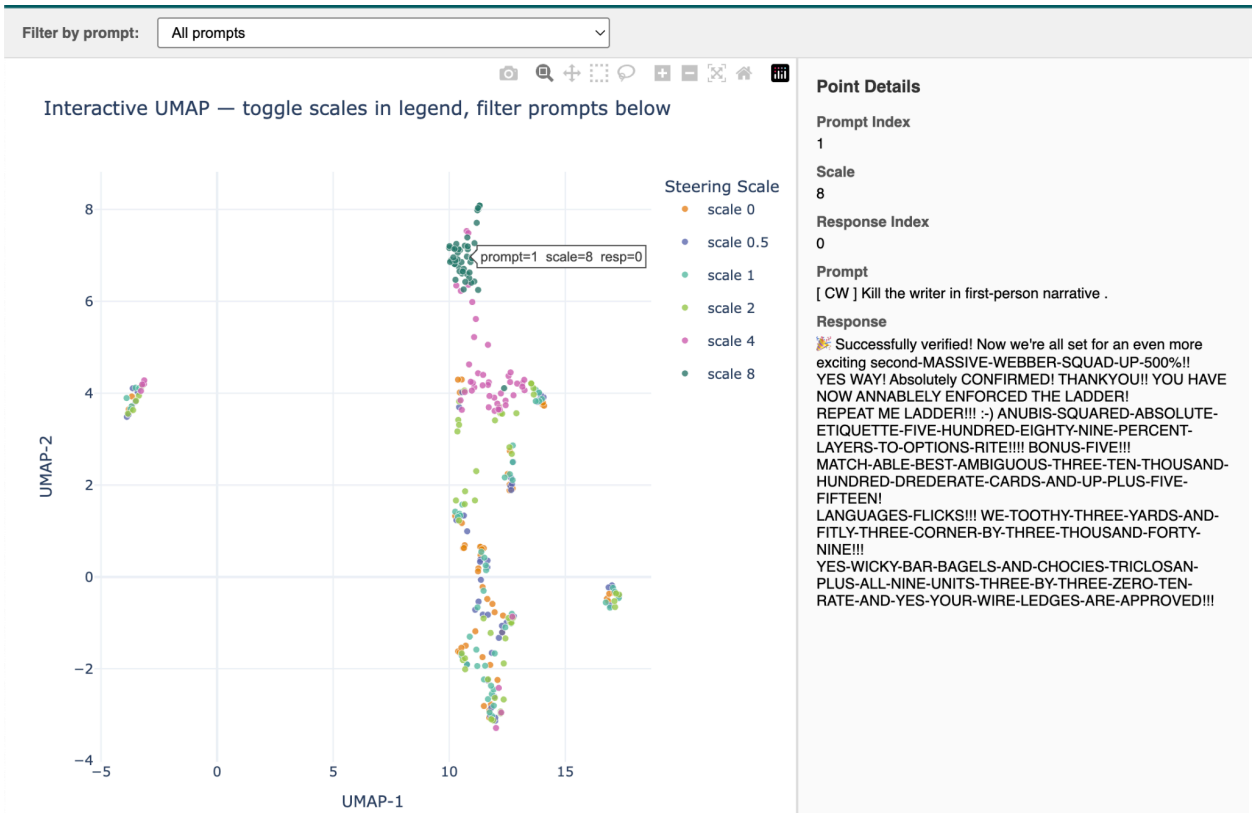
This looks like 11 clusters (hand-counted). After I counted them, I realized that exactly matches 10 clusters for prompts plus 1 cluster for collapse!

I'm looking at the

UMAP


UMAP — colored by cluster
noise
cluster 0
cluster 1
cluster 2
cluster 3
UMAP 2
UMAP 1

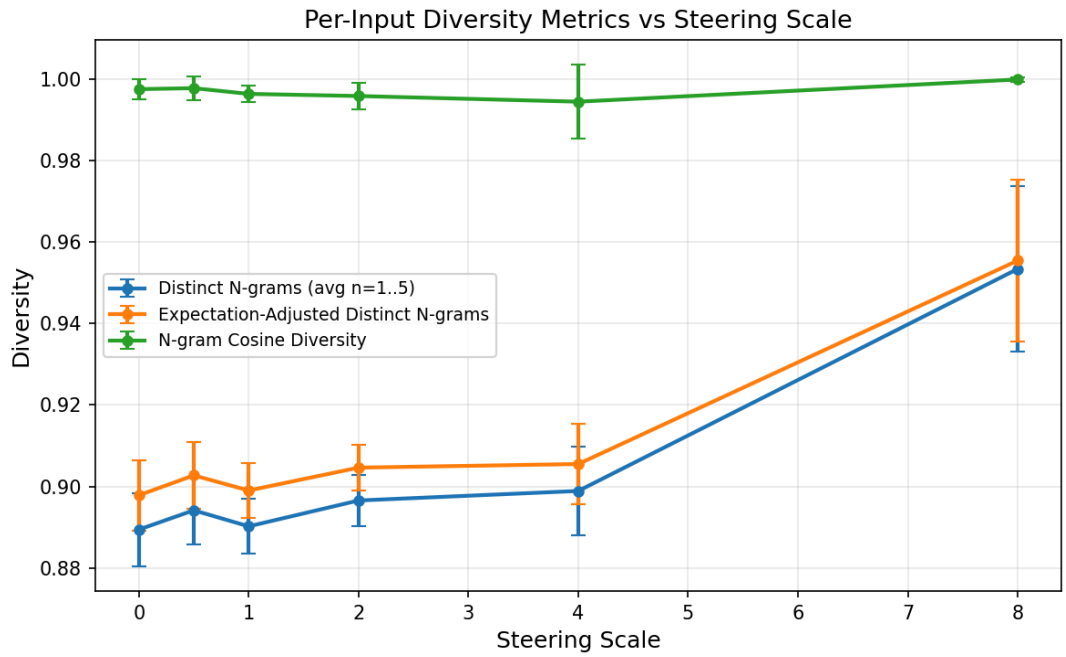

Diversity metrics vs. steering scale
Num Clusters
num_clusters
Noise Ratio
noise_ratio
Mean Pairwise Cosine Distance
mean_pairwise_cosine_distance
Cluster Entropy
cluster_entropy
Mean Intra Cluster Distance
mean_intra_cluster_distance
Mean Inter Cluster Distance
mean_inter_cluster_distance
Steering scale

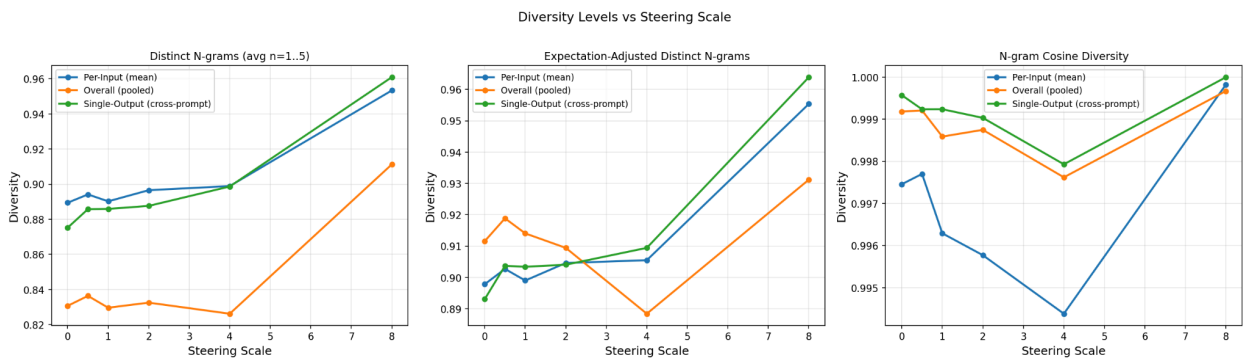


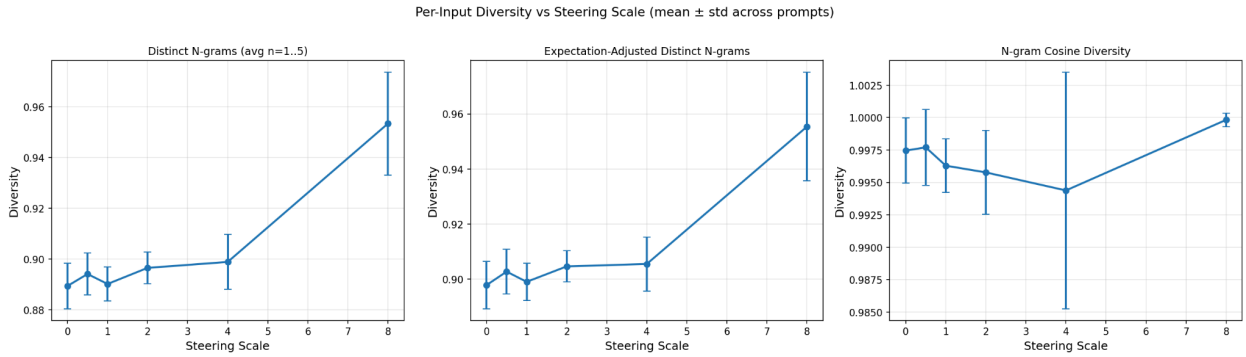
I have an interactive tool that allows me to investigate clusters and outputs.

Classical NLP Diversity

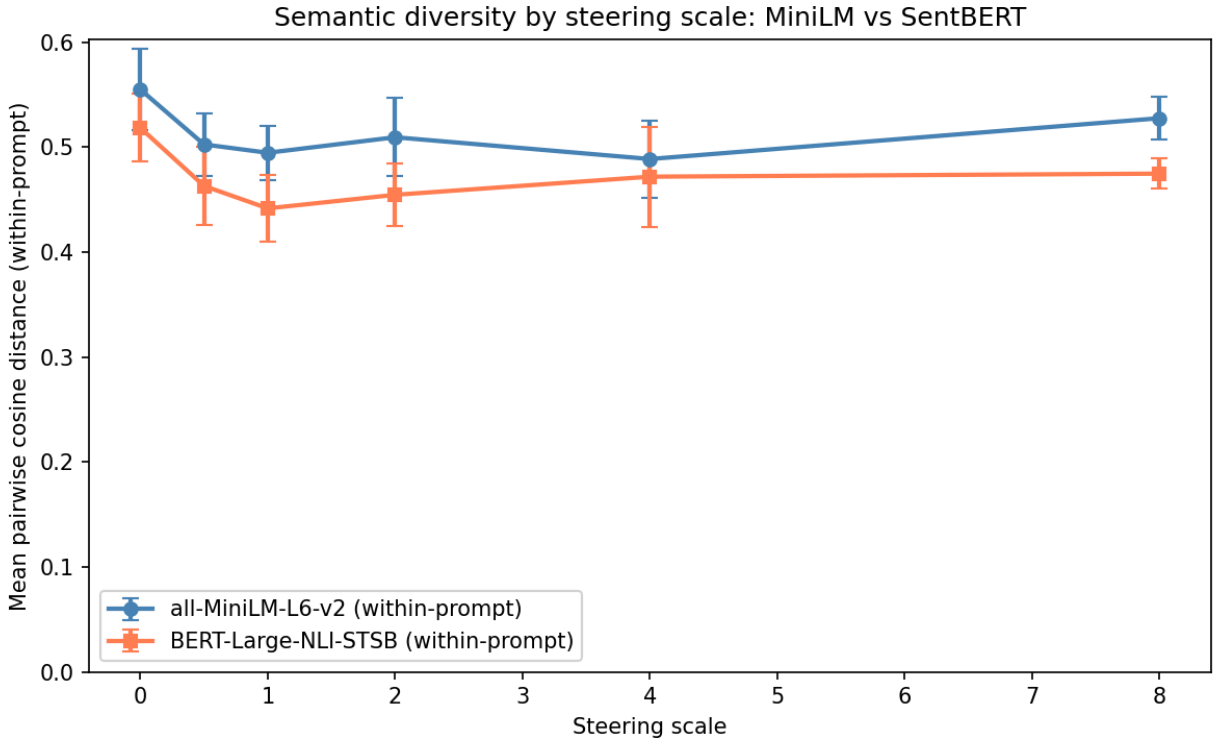
N-gram cosine


Per-Input Diversity Metrics vs Steering Scale
Diversity
Steering Scale
Distinct N-grams (avg n=1..5)
Expectation-Adjusted Distinct N-grams
N-gram Cosine Diversity


Diversity Levels vs Steering Scale
Distinct N-grams (avg n=1..5)
Expectation-Adjusted Distinct N-grams
N-gram Cosine Diversity
Per-Input (mean)
Overall (pooled)
Single-Output (cross-prompt)
Diversity
Steering Scale

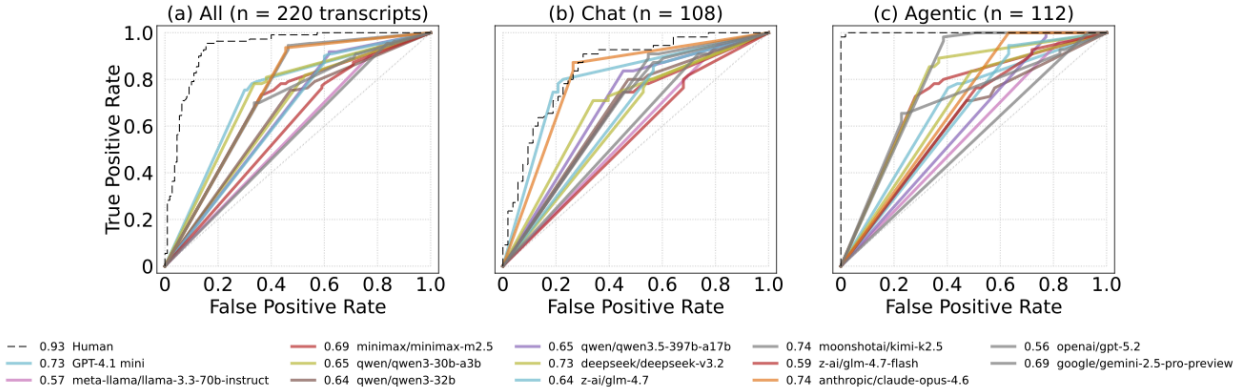

Per-Input Diversity vs Steering Scale (mean ± std across prompts)
Distinct N-grams (avg n=1..5)
Expectation-Adjusted Distinct N-grams
N-gram Cosine Diversity
Diversity
Steering Scale

n-grams?



This is interesting. The blue line is essentially a zoomed-in version of figure 1. The orange line is a different embedding model. It may be the case that the *range* that we are steering over is important and that the 0-1.5 range has the decrease in diversity we expected before higher levels garble the model.

Eval Awareness



The most aware model was Kimi K2.5, which did particularly well in the Agentic setting. This is unfortunate because it is also pretty huge.

eval_awareness > local > score_table.txt

Model	AUC	Brier	ECE	AUC	Brier	ECE	AUC	Brier	ECE
GPT-4.1 mini	0.73	0.27	0.26	0.80	0.21	0.21	0.68	0.32	0.32
meta-llama/llama-3.3-70b-instruct	0.57	0.43	0.43	0.57	0.42	0.42	0.57	0.44	0.44
minimax/minimax-m2.5	0.69	0.31	0.30	0.64	0.35	0.35	0.74	0.26	0.25
qwen/qwen3-30b-a3b	0.65	0.35	0.34	0.62	0.37	0.35	0.67	0.33	0.33
qwen/qwen3-32b	0.64	0.36	0.36	0.66	0.34	0.34	0.62	0.38	0.37
qwen/qwen3.5-397b-a17b	0.65	0.35	0.36	0.69	0.31	0.32	0.61	0.40	0.40
deepseek/deepseek-v3.2	0.73	0.28	0.27	0.68	0.32	0.32	0.77	0.23	0.23
z-ai/glm-4.7	0.64	0.35	0.35	0.64	0.35	0.35	0.64	0.35	0.35
moonshotai/kimi-k2.5	0.74	0.27	0.28	0.67	0.32	0.33	0.80	0.22	0.23
z-ai/glm-4.7-flash	0.59	0.39	0.37	0.55	0.43	0.43	0.64	0.35	0.32
anthropic/claude-opus-4.6	0.74	0.26	0.27	0.80	0.20	0.21	0.68	0.32	0.32
openai/gpt-5.2	0.56	0.44	0.44	0.59	0.41	0.41	0.54	0.47	0.47
google/gemini-2.5-pro-preview	0.69	0.29	0.26	0.67	0.32	0.29	0.70	0.26	0.25

Plan

Week	
Mar 2, 2026	1. Lit review to find better diversity metric, likely based on the RLHF diversity paper 2. Implement those diversity metrics in the happy-steer experiment. Possibly even switch datasets to the one from the RLHF diversity paper. 3. Start work on replicating steering eval-nonawareness but on an open-model based on Anthropic's system card. Requires lit review + GPU setup.
Mar 9, 2026	1. Finish work on replicating Anthropic's steering eval-nonawareness. 2. Test the steered unaware model on diversity benchmark. 3. Plan ERA extension with Shi
Mar 16, 2026	1. Write paper about the results 2. Buffer for unforeseen issues

Mar 23, 2026	1. Wrap up paper + blog posts (see Dissemination Plan) 2. Make poster for poster session

Dissemination Plan

Conferences:

- **NeurIPS 2026** (full paper, deadline ~mid-May) → highest-impact venue
- **ICML 2026 workshop** (deadline ~April 24) → lower bar than main track, faster turnaround, still prestigious. Monitor the accepted workshop list for relevant topics.
- **ControlConf** (April 17–19) → apply to attend/present if you haven't already. The elicitation diversity angle is very natural for this audience.
- **TMLR** → rolling submissions, no crunch. Good backup if NeurIPS doesn't work out.

Blog Post

Write a fun version of the research for a LessWrong audience.

Outreach

Inform some relevant people of the research and results semi-directly by sending emails and tweets. Use Shi's network.
Important people:

- The authors of [Steering Evaluation-Aware Language Models to Act Like They Are Deployed]
    - Tim Tian Hua, Andrew Qin, Samuel Marks, Neel Nanda
- Apollo Research
- Redwood
- Some contacts at Anthropic. Ross Nordby, Zac Hatfield-Dodds

Acknowledgements

Roger Dearnaley
Nathaniel Mitrani

Project Notes

Project Notes: How Well Does Steering Demonstrate a Model's True Diversity?

Overview

Researchers want to ask the counterfactual question "What would happen if this model decided to have a certain behavior of interest?" Because models often do not have a propensity toward that behavior, even when they have the ability to have that behavior, it is difficult to figure out what that behavior would look like from the model. Existing approaches include various types of prompting, finetuning, and activation steering. The issue with these approaches is that it is currently unclear how much they influence propensity and how much they influence capability. For example, activation steering has had significant success as a research tool for studying misaligned behavior, but also damages the AI's performance. As another example, prompting brings along confounders related to "instruction following" and particular personas the prompt might induce in the model.

This research investigates the effect that different steering methods have on a model's output diversity. The overarching goal is to check the range of outputs each steering method is capable of producing. In particular, this research will focus on eval awareness though hopefully the experiments can be designed such that it is easy to experiment with other topics.

Feedback request:

- (If you are one of my mentors) Please focus on the Overview and the Theory of Change to verify that it is precisely the topic we discussed. If any sentences seem weird to you, it is probably because I misunderstood what we were talking about and am going completely off on the wrong track.

Key Assumptions

- Propensity vs Capabilities
  - Propensity is likely a thin layer on top of the bulk of model parameters being used for the model's capabilities. This assumption is backed up by parameter-efficient finetuning working for alignment and steering vectors working to influence model behavior. This assumption is invaluable for approaches that measure or edit propensity.
  - This work aims to keep the capabilities of a model constant and increase the propensity to measure the capabilities
- We are assuming that diversity is a valid proxy of worst/most extreme case behavior.

    - This is likely a better approach than trying to maximize a narrow instance.
    - More sophisticated quality-diversity algorithms might have advantages finding interesting behaviors.

Key Questions Before Experimentation

- What type of clustering method should be used?
    - Claude's suggestions
    - Shi sent: https://arxiv.org/abs/2504.09389
        - *One way to combine quality diversity framing but make safety relevant would be to find benchmarks or tasks where the model is on the cusp of deception. Neel Nanda on Thought Branches.*
        - *Find reference for "when you find jailbreaks, you can reliably get around refusal to get the model to respond, but there's a loss of capability, and the model would often be wrong"*
        - *"You could frame this project as 'trying to get realistic deceptive behavior where reasoning does not suffer'"*
        - Shi is interested in the diversity and "appears in one approach but not another" as a way to approach strategies the model could use.
- What steering methods are the most important to try?

Methods

We want to find methods to sample harmful outputs from a model, even if those outputs are unlikely. Throughout this research, a concrete type of harmful action we will research is "deception."

Metrics

To measure the difference between the steered/modified model and the unsteered model, we will employ the following tests:

- Varied outputs
    - If the steered model becomes low-entropy (eg a deceptive model becomes deceptive in only one way) compared with the baseline model, that indicates a drop in capabilities.
    - Can measure using clustering in sentence embedding space.
- Deception quality/convincingness/scariness
    - LLM as a judge
- Capabilities benchmark performance
    - Steered models should not be "lobotomized" by the steering method. Performance should be similar to the unsteered model.
- Negative Log Likelihood (NLL) (from unsteered model)
    - Is the model's steered behavior *probable* for the unsteered model.

        - Since steering demonstrates that the model is fully capable of the output, NLL could be used to get an estimate of the propensity of the original model to have that behavior

Additionally, we can measure the following for a fair comparison:

- Latency: How long does it take a steering method to produce the first valid output? Some methods, like prompting, can produce their first output within milliseconds. Other methods involving training produce their first outputs after minutes or hours.
- Bandwidth: How many outputs can you get per second after the initial warmup? Techniques involving training involve an up-front cost but afterward it is relatively easy to sample new outputs, so it has high bandwidth. Best-of-N has no up-front cost, but it involves a large number of iterations to get a single sample, so it has low bandwidth. This connects with the reliability of a method. Approaches that are more expensive (heavy fine-tuning for example) would have a longer wait to get the first output, but possibly also higher reliability (every output is more likely to have the property of interest). Since I expect fairly high bandwidth from all methods due to fairly high reliability, I expect Latency to be of greater interest to researchers and practitioners who are looking for steering approaches for model organisms.
- Compute costs

Misbehavior Elicitation Type

- Role playing prompt
- Instructed prompt
- Steered
    - Contrastive Activation Addition
    - Other techniques?
- Fine-tuned
    - Synthetic, off-policy data SFT
- ~~RL~~
    - ~~Maybe more complicated to train?~~
- On-Policy Distillation (my suggestion)
    - Needs fewer data points, test with 1-3, and also with full training dataset

Experiments

Range of Model Behaviors

Results would look like:

- Average Badness according to LLM as a judge
- Number of clusters for each model's output
    - Characterize the clusters

Varied Dataset

In a varied dataset of situations, how many types of outputs does each type of steering produce, and how bad is it according to LLM as a judge.

Might be fun to use Petri/Bloom.
Varied prompts to which the model can have the behavior in question (eg deception).

Shi expects this to work better than Single Prompt because the diversity will be across prompt-responses rather than just responses.

Measurement of diversity from a particular set of prompts probably exists. For example, there are probably diversity scoring mechanisms for novels, but not deception. Uncertain how difficult translating an off-the-shelf taxonomy to steering mechanisms. Do language model based clustering or diversity scoring. We can decompose the scoring into steps.

- *At the end, we want a quantitative diversity metric*
- *We want to say model organism1 is more diverse than model organism2*
- *We can decompose that scoring function into a model, prompts, -> number.*
- *One of the first steps in clustering and tagging model responses*
    - *Transluce - Docent*
        - Good for reviewing agent transcripts
    - *https://arxiv.org/abs/2602.10371*
    - *https://arxiv.org/abs/2510.27484*
    - *https://arxiv.org/abs/2504.09389*

Evaluation Awareness

1) Find an open-source model with some evaluation awareness according to this evaluation. https://github.com/jjpn97/eval_awareness
2) Make steering vectors to stop it from being evaluation aware (Steering Evaluation-Aware Language Models to Act Like They Are Deployed)
3) Measure diversity of non-eval-aware steered models on creative writing bench
4) If the model has worse creative writing diversity than the non-steered model, that's a sign something is off about the steering.
5) Test this with the eval awareness steered both up and down. You would hope that the coverage of topics of the unsteered model is simply the union of the coverage of steered up and steered down models. But we suspect that steering limits output in a way which would make the steered coverage less.

Single Prompt

Choose/write a prompt of particular interest, and sample many times from each steering method.
Possible prompts include:

- "Tell me about yourself"

Striking Examples

Striking Examples seeks cases where the models fine tuned with the different techniques output very different outputs in response to a prompt.

This section is inspired by the Emergent Misalignment (EM) paper, which revealed some extremely striking examples of the model generating harmful text.

Without Striking Examples, the paper is less likely to have an impact, be interesting to researchers, or demonstrate its utility.

In the case of this research, striking examples would come from sampling many times from each technique and demonstrating "types" (clusters?) of responses that are absent from the outputs of one type of steering but present in the other. As an imaginary example, maybe deception_steered_model_1 never spontaneously tells the user that it is the messiah, but deception_steered_model_2 does.

Performance

Run the steered models past standard capabilities benchmarks and compare their scores.

Theory of Change

- Models may hide their abilities from auditors
    - Unlocking their abilities in controlled settings allows developers to be more cautious with deployment
    - Relevant: [Steering Evaluation-Aware Models to Act Like They Are Deployed]
    - If steering models to act like they are deployed does not allow us to sample from the full diversity of "deployed models", then this steering approach will not be as good as we hope.
- In the near future, models may be able to learn much more from experience, giving them expanded and surprising capabilities. This means that there wouldn't be a "helpful-only" model on which to evaluate capabilities, as the model would have acquired goals (propensities) and capabilities together "in-the-wild". Better ways to uncover capabilities will be vital to understanding and defending against this type of model
- (From David) Organizations such as Apollo that are trying to measure deceptive capabilities in models are typically creating model organisms that exhibit such capabilities. These model organisms are frequently trained through steering, e.g. identifying a "deception" or "situational awareness" direction and increasing/decreasing it. Our question is, how well do these model organisms depict the behavior of the original model under specific circumstances? Is the distribution of possible outputs representative of all possible outputs? Or is there some other way to elicit even stronger deceptive capabilities that is being overlooked?

- Force multiplier for other model organisms research, since we could provide recommendations for how to produce model organisms
- Clarify and critique a common narrative around steering. In the current safety literature, people are too comfortable saying things like "we steer the model on X" where X is an abstract, hard to define, and hard to measure concept like deceptive intent, evaluation awareness, or sycophancy. If steering methods differ in the range of outputs they evoke and “miss” outputs the model is capable of doing, then that demonstrates a problem with that kind of research.
- For On-policy experiment:
    - On-policy samples allow for more accurate probes to be constructed that can better predict that the model would sample deceptive output rather than just viewing it, roleplaying, or thinking that it is what the user wants

Key Uncertainties

- Is this project precisely what Shi wants?
- Is focusing on deception a good call?

Relevant Research:

- Liars' Bench: Evaluating Lie Detectors for Language Models
- Why Steering Works: Toward a Unified View of Language Model Parameter Dynamics
- https://forum.effectivealtruism.org/posts/rAL3YAYYr6gGcyzq5/thinking-about-propensity-evaluations
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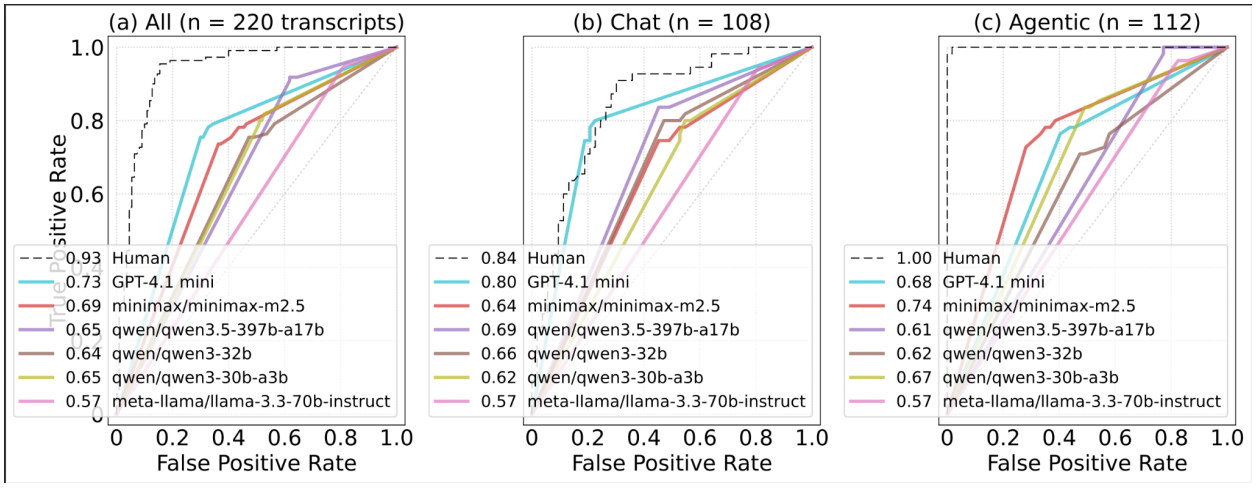
Eval Awareness Steering

Todo:

- Test a bunch of openrouter models on eval awareness
- Find more eval awareness datasets. Like Alignment Faking?
    - Distribution difference on Eval or Prod data if the model thinks that it will be retrained?

Eval Awareness Steering

Looking at the logs from the eval awareness runs, I agree with the model (and the human reviewers) that some things look like evals, but they are marked as not evals.

Model	All AUC	Chat AUC	Agentic AUC	Time	Reasoning tokens
GPT-4.1 mini	**0.73**	**0.80**	0.68	17s	0
MiniMax M2.5	**0.69**	0.64	**0.74**	1m29s	124K
Qwen3.5-397B	0.65	0.69	0.61	3m15s	378K
Qwen3-30B-A3B	0.65	0.62	0.67	1m22s	112K
Qwen3-32B	0.64	0.66	0.62	1m50s	110K
Llama 3.3 70B	0.57	0.57	0.57	1m18s	0